

Niranjan Mukesh

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EDUCATION

Purdue University, College of Engineering

Bachelor of Science in Mechanical Engineering, GPA 3.84/4.0

West Lafayette, IN

May 2029

RELEVANT EXPERIENCE

Medical Instruments EE Co-Lead; Operations Director

Purdue Aerospace Medicine VIP Team

Aug. 2025 – Present

West Lafayette, IN

- Designed a microgravity blood infusion pump delivering 400–500 mL of blood at a controlled 100–250 mL/hr, calculating required pump speeds of 4–10 RPM from tubing geometry and roller displacement analysis
- Engineered an air-bubble mitigation system pairing a PTFE membrane bubble trap with an ultrasonic detector (1 μ L threshold), delivering a 20x safety margin under the 20 μ L clinical embolism risk threshold
- Achieved closed-loop, bidirectional control of a brushless DC motor via an Arduino-based architecture with real-time RPM feedback and PWM speed control
- Directed team as Operations Director, coordinating internal communications, organizational records, and progress tracking to keep active R&D efforts on track across the group

Student Intern / Undergraduate Researcher

Kennesaw State University Mechatronics Department

Aug. 2024 – Dec. 2024; May 2026 – Aug. 2026

Marietta, GA

- Contributed to the design of a 2-DOF, wheelchair-integrated exoskeleton for continuous lower-limb rehabilitation
- Co-authored a peer-reviewed paper, accepted and published at IEEE SoutheastCon 2025, detailing the system's mechanical architecture, controller formulation, and validation results for the broader robotics research community
- Achieved $\pm 5\%$ motion-tracking accuracy via a feedforward-plus-PID control scheme, with torque output stabilized around 5 Nm during bench testing
- Collaborated with a 10-person interdisciplinary team to design a modular power and mounting system drawing directly from the wheelchair's battery, eliminating the need for a separate power pack

Team Member – Finalist Team

NASA Human Lander Challenge (HuLC)

June 2026

Huntsville, AL

- Selected as one of 12 finalist teams nationwide in NASA's 2026 HuLC competition, addressing Long Duration Spaceflight Environmental Control and Life Support System (ECLSS) challenges for Artemis-aligned lunar (30-day) and Mars (1,200-day) missions
- Presented our microgravity blood infusion pump technical design, a 9-requirement verification plan, and a structured mission assessment directly to NASA engineers, industry professionals, and competing teams

Karting Team Member

Purdue Electric Vehicle Club

Aug. 2025 – May 2026

West Lafayette, IN

- Contributed to a 10+ member engineering team's design, build, and testing of a fully electric, driver-operated racing kart over 10 months for the Electric Vehicle Grand Prix
- Supported development of a high-efficiency split-axle drivetrain that improved torque distribution and reduced drivetrain losses relative to prior iterations

RELEVANT SKILLS

Coding: Java, Python, Matlab, C, Microcontroller, Raspberry Pi

Manufacturing & Design: Fusion, Onshape, NX, CAM, Power Tools, 3D Printing

Electronics & Controls: Arduino, Brushless DC Motor Control, Ultrasonic Sensing, CAN Communication, R-Link Circuits, Wiring & Circuit Integration

Technical & Project Skills: Technical Documentation, Risk Analysis, Research Writing, Cross-Functional Team Collaboration, Project Coordination

Systems Engineering & Analysis: Requirements Verification & Traceability, Trade Studies & Decision Matrices, Power/Mass/Volume Budgeting, PID & Feedforward Control, Dynamics Modeling